source ..... noise engineering manual
rev 2026.10

imitor versio

_travelers guide
_for chasing echoes

12	96k	7
taps	sample rate	knobs · 2 switches · tap

[fsu] press **[fsu]** hold › then noon = 12 o'clock numbers = the numbered panel

12-tap multimode stereo delay · tap tempo + clock sync · 96 kHz · a free firmware for any versio

← back to the versio starsystem

cyan = hold

01 what it is

iv_01

a delay	twelve taps, multimode, stereo in and out · runs free, from tap tempo or from a clock
in 1 in 2	stereo in · in 1 alone feeds both sides
out 1 out 2	stereo out
every knob	an offset added to its cv jack, 0 – 5 V
swap in	free on any versio · how to swap

02 blend · regen

iv_02 · knobs 1 + 5

1 [blend]	dry input to delay
5 [regen]	feedback · keep it low while learning
past 3:00	the delay ducks under the input, sidechain style
in shm	any regen adds pitch shift, plainest past noon

03 time · spread

iv_03 · knobs 6 + 7 · fig.01

7 [spread]	base delay time: from now to the last tap
6 [time]	wait before the first tap, as a share of spread
clocked	with tap or a clock, spread divides or multiplies it
cv on spread	an lfo here gives the big warbles and pitch bends

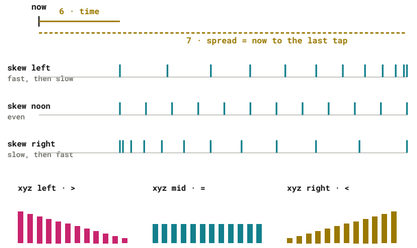
10 first flight

iv_10 · a clean start

1. sound into in 1 (and in 2 for stereo) · out 1 out 2 to your mixer
2. 1 [blend] at noon · 5 [regen] low
3. 2 [skew] + 3 [angle] at noon · abc lim · xyz = (mid)
4. set the length with 7 [spread], then the first tap with 6 [time]
5. tap [fsu] or patch a clock to lock it to the beat
6. now move skew, angle and the lfo · bring regen up last

_fig.01 : twelve taps

time · spread · skew · xyz



04 skew · angle

iv_04 · knobs 2 + 3 · fig.01 · fig.02

2 [skew]	tap spacing · noon even · left starts fast and slows · right starts slow and speeds up
3 [angle]	turns each tap around the stereo circle from the last one
angle 1:00	taps drift right, step by step, then wrap round to the left
angle far	extreme settings bounce taps all over the field

_fig.02 : angle

where each tap sits in the stereo field

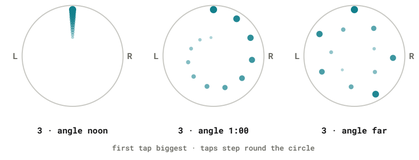
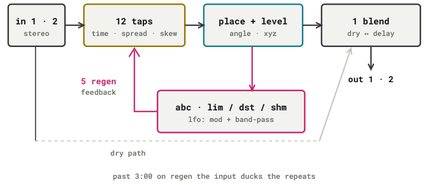


_fig.05 : signal path

as the manual describes it



past 3:00 on regen the input ducks the repeats

05 lfo

iv_05 · knob 4 · fig.04

4 [lfo]	modulation plus a wide band-pass on the delay
left of noon	random modulation
right of noon	triangle modulation
tone	it changes the colour of the repeats, not just their wobble

_fig.04 : lfo

plus a wide band-pass



06 switches

iv_06 · abc · xyz

abc left	[lim] clean feedback
abc mid	[dst] slightly distorted feedback
abc right	[shm] pitch-shifted shimmer feedback
xyz left	[>] decrescendo · each tap quieter
xyz mid	[=] all taps equal
xyz right	[<] crescendo · each tap louder

08 six patches

iv_08 · from the manual · abc · xyz

angry duck — lim or dst · <
short, low sound into in 1 · regen fully up so the input ducks the delay

snap crackle pop — shm · =
a clock or trigger pattern into in 1 · different clocks into each in for width

bouncing ball — lim · >
short plucky sound in · move angle to throw the bounce around

reverse delay — lim · <
anything in · crescendo plus skew reads as a reversed echo

shimmers — shm · >
longer melodic sound · regen above noon for the pitch shift

80s — dst · >
anything in · lfo for colour · cv into spread for wild warble

07 tap · clock

iv_07 · fsu

[fsu]	tap tempo · the LEDs flash purple on each tap or clock
fsu jack	clock in · gates above 2 V
[fsu] 2 s	forget the tapped tempo · LEDs flash orange · knobs take over
[fsu] 5 s	empty the delay lines · LEDs flash white
boot	cyan · cyan · yellow · white = imitor is loaded

_fig.06 : what the LEDs say

all four lights

boot	● ● ● ○	imitor is loaded
each tap / clock	● ● ● ●	flash on every pulse
hold fsu 2 s	● ● ● ●	tapped tempo forgotten
hold fsu 5 s	○ ○ ○ ○	delay lines emptied

_fig.03 : the panel

numbers match the legend



1. **blend** · dry ↔ delay
2. **skew** · tap spacing
3. **angle** · stereo turn per tap
4. **lfo** · random ← · → triangle
5. **regen** · feedback · ducks past 3:00
6. **time** · wait to the first tap
7. **spread** · now to the last tap

abc **feedback** · lim · dst · shm
xyz **tap levels** · > fade · = level · < grow
fsu **tap** · tap · clock in · holds
in **in 1 / 2** · in 1 feeds both if alone
out **out 1 / 2** · stereo out

09 specs

iv_09

cv in	0 – 5 V
fsu gate	above 2 V
audio in	clips near 16 V peak to peak
rate	96 kHz
size	10 hp · 1.5 in deep
power	version a 70 mA +12 V · 70 mA –12 V · version b 125 mA · 10 mA · no 5 V

design homage to the house of x / powers of x data pages — designed by tom muller with jonathan hickman · panel art: noise engineering
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